

7.

Ans: C

$$\begin{aligned}\text{Energy} &= \frac{1}{2}mv^2 \\ &= \frac{1}{2}m(u + at)^2 \\ &= \frac{1}{2}m(a^2)(t^2) \quad \text{since } u \text{ is zero} \\ &= \text{constant} \times t^2\end{aligned}$$

8

Ans: D